MIL-N-87963(USAF)
27 March 1979
SUPERSEDING
AF 82-PD368
25 March 1976

MILITARY SPECIFICATION

NOZZLE: FUEL, AIRCRAFT, OVERWING: TYPE MD-3

This specification is approved for use by HQ AFLC CASO/LODS, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one kind of shockproof, overwing, aircraft, fuel servicing nozzle, designated Type MD-3.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-T-291	Thinner, Paint, Volatile Spirits (Petroleum-Spirits)

Military

MIL-H-775	Hose, Rubber, Plastic, Fabric, (Including Tubing): And Fittings Nozzles and Strainers, Packaging of
MIL-G-1086	Gasket Material, Synthetic Rubber (For Bolted Steel Tanks)
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-R-25897	Rubber, Fluorocarbon Elastomer, High-Temperature Fuel Resistant
MIL-I-27686	Inhibitor, Fuel System Icing

STANDARDS

Military

MIL-STD-130	Identification Markings of US Military Property

FSC 4930

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MIL-STD-266	Impregnation of Porous, Nonferrous Metal Castings
MIL-STD-810	Environmental Test Method
MIL-STD-831	Test Reports, Preparation of
MS25384	Electrostatic Discharger, Jumper, Fuel Nozzle to Aircraft

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw Thread Standards For Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing.

3.2 Components. The nozzle shall consist essentially of the following components:

	Component	Requirement
a.	Body	3.7.1
b.	Valve	3.7.2
c.	Handle and operating lever	3.7.3
d.	Spout	3.7.4
e.	Strainer	3.7.5
f.	Filler neck protector	3.7.6
g.	Dust cap	3.7.7
h.	Electrostatic Discharger	3.7.8

3.3 Materials. Nozzle materials shall be as follows:

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3.3.1 Metals. Metals that normally come in contact with the dispensed fuel shall be, except for the special requirements listed below, restricted to wrought aluminum alloys in the 1000, 3000, 5000 and 6000 series; cast aluminum alloy 356 and 300 series stainless steels. Nozzle spout shall be fabricated from 6061-T6 aluminum. Strainer joining material may be any metallic solder or brazing material that does not contain either magnesium or copper except as impurities. Strainer, valve stem and valve spring shall be fabricated from 300 series stainless steel. Metals not normally in contact with the dispensed fuel shall be either aluminum alloy or stainless steel except that small metal parts (bolts, screws, nuts, washers, et cetera) shall be either cadmium or zinc plated steel. This document imposes no restrictions on the electrostatic discharger.

3.3.2 Metal treatment. All aluminum alloy parts shall be anodized in accordance with MIL-A-8625, Class I, either Type I or Type II. All aluminum casting that are subjected to fluid pressure in service shall be impregnated in accordance with MIL-STD-266. All thread surfaces on aluminum alloy parts shall be treated with solid film lubricant shall be in accordance with the applicable provisions of the Appendix to MIL-L-46010.

3.3.3 Nonmetals. Internal elastomers, gaskets and seals, including stem seals, shall be fabricated from either rubber conforming to MIL-R-25897, Type and Class as applicable, or commercial grade polytetra-fluoroethylene, or commercial grade asbestos impregnated with polytetra-fluoroethylene. Nozzle spout dust cap shall be molded from rubber conforming to MIL-C-1086, Type II. Filler neck protector bumper shall be of any commercial, aromatic fuel resistant, elastromeric material.

3.4 Design. The nozzle shall be suitable for open-flow dispensing of aviation gasoline, jet fuel and water-alcohol mixture (ADI) to aircraft fuel tanks. The nozzle shall be of the balanced-pressure self-closing type and shall be designed so that the force required to open the valve and control the flow remains practically constant independent of the flow rate. The nozzle shall not include any automatic shutoff device nor any means of locking in open position. The nozzle shall conform to Figure I in dimensions and general configuration.

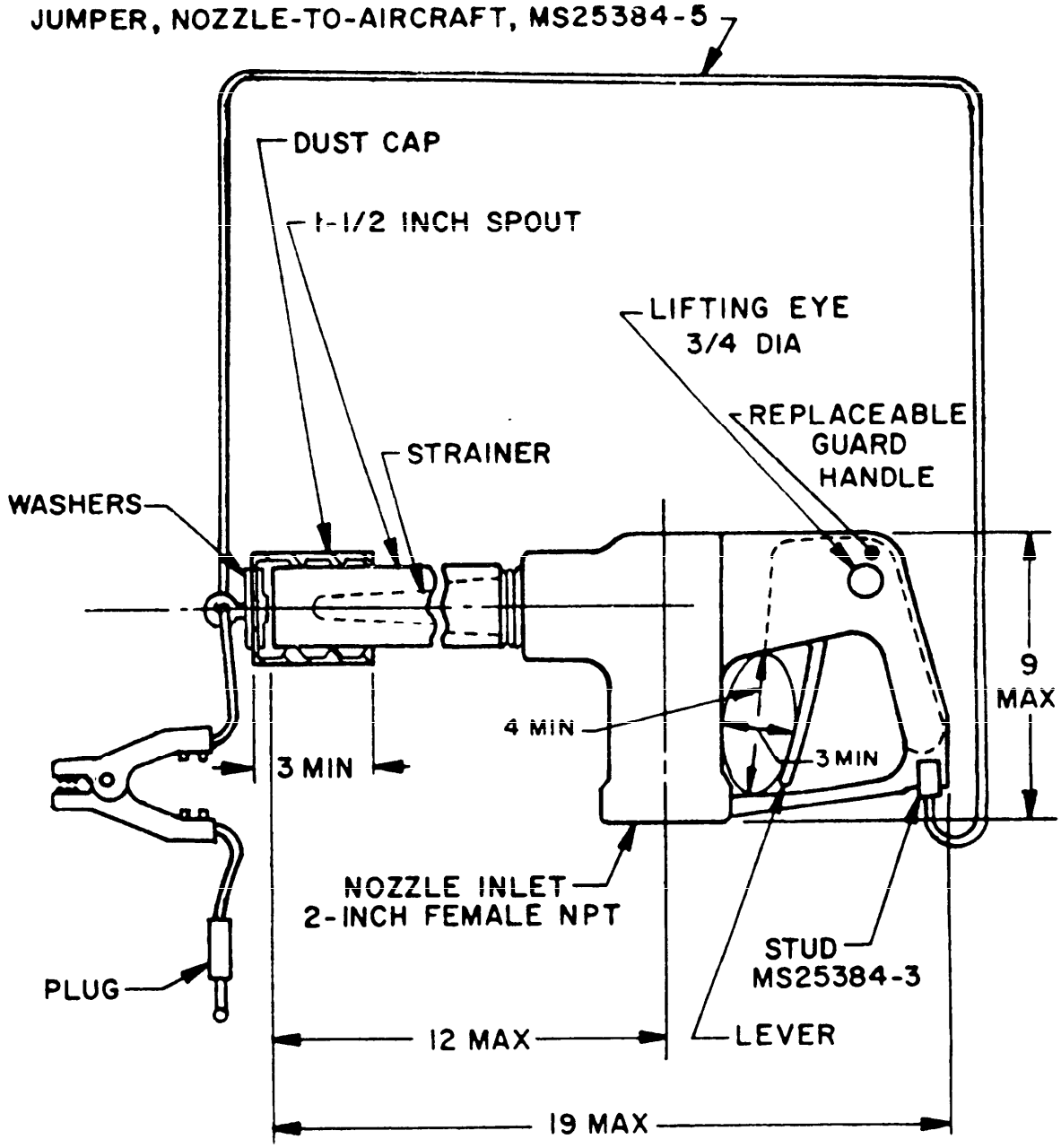
3.5 Construction. The nozzle shall be constructed so that no parts will work loose in service. It shall withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The nozzle shall be constructed so that adjustments and repairs can be made easily by personnel of operating units and overhaul bases without the use of special tools.

3.6 Performance. The nozzle shall have the following performance characteristics when tested as specified herein.

3.6.1 Capacity. The nozzle flow capacity shall be not less than 135 gallons per minute (gpm) at a nozzle inlet pressure not greater than 25 pounds per square inch gage (psig).

3.6.2 Closing. The nozzle shall be self closing and shall shut off fuel flow within 1/5 seconds after release of the operating lever.

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DIMENSIONS IN INCHES

TOLERANCE:FRACTIONS ± 1/8

FUEL, NOZZLE, TYPE MD-3
FIGURE 1

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3.6.3 Surge. Surge pressure due to release of operating lever shall not exceed 50 pounds per square inch (psi).

3.6.4 Leakage. The nozzle shall not leak or weep when tested as specified herein.

3.6.5 Lever pull. Lever pull force to operate the nozzle shall not exceed 15 pounds.

3.6.6 Impact. The nozzle shall withstand multiple 12 foot drops onto a concrete surface.

3.6.7 Environmental. The nozzle shall meet the requirements specified herein during and after exposure to the following:

(1) Temperature: Operating, -65°F to 125°F; storage, -65°F to 150°F.

(2) Salt spray: Exposure to salt-laden atmosphere.

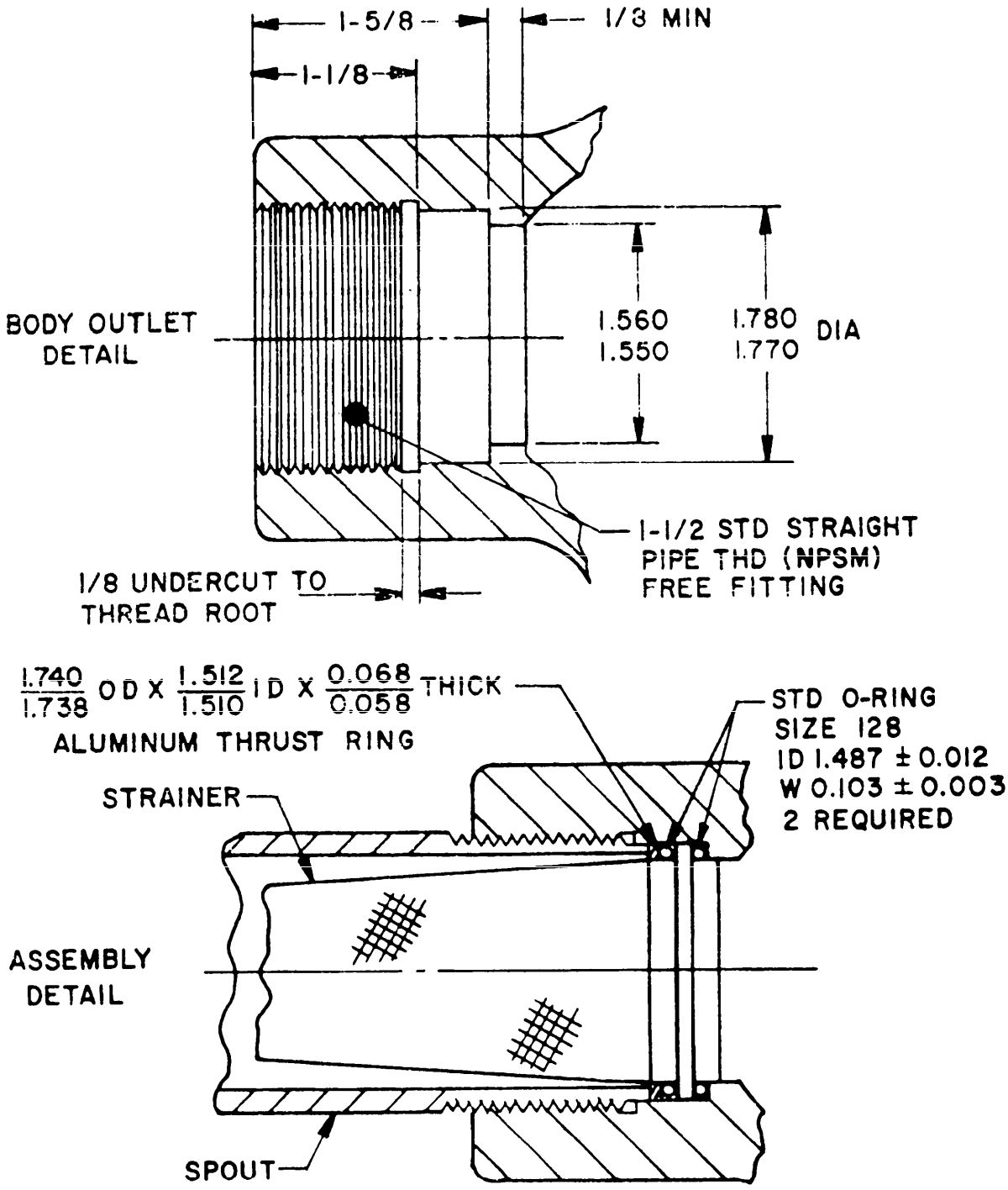
3.7 Details of components.

3.7.1 Body. The nozzle body shall conform to Figure 1 in dimensions and general configuration, with a 90 degree angle between centerlines of inlet and outlet connections. The nozzle body shall be fabricated separate from the guardhandle. The outlet connection of the nozzle body shall conform to Figure 2. Wear lugs of wear rings (or both lugs and rings, if necessary) shall be provided to prevent abrasion of vital parts when the nozzle is dragged. The wall of the nozzle body shall be not less than 3/16 inch thick. After heat treatment and completion of all machining operations, the body shall be successively anodized, impregnated, and then treated with solid-film lubricant on the threated surfaces; all of these treatments being performed in accordance with 3.3.2.

3.7.2 Valve. The nozzle valve shall be designed and constructed to shut off the fuel flow within not more than 1.5 second after release of the operating lever and without surge pressure in excess of 50 psi. The valve closing time shall be fixed by design; i.e., no adjustable device is permitted. The valve stem shall be furnished with an adjustable packing gland or a readily replaceable O-ring seal or seals to prevent leakage.

3.7.3 Handle and operating lever. The nozzle shall be equipped with a handle securely attached to the nozzle body in the general configuration shown in Figure 1. The handle shall be readily removable and replaceable by means of ordinary tools. The handle shall include a 3/4-inch diameter lifting eye in the general location shown in Figure 1; this eye shall be an integral part of the handle and shall not protrude beyond the main contour of the handle. The handle shall be equipped with an operating lever that actuates the nozzle valve; release of the lever shall cause closure of the valve. The handle shall function as a lever guard and shall enclose the lever completely in such a manner as to protect the lever from damage when the nozzle is dropped. The handle shall not restrict or interfere with free motion of the lever in its normal plane of motion; no notches or any other locking or arresting device shall be

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DIMENSIONS IN INCHES TOLERANCE: FRACTIONS ± 1/32

NOZZLE BODY OUTLET AND ASSEMBLY DETAIL

FIGURE 2

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permitted. The lever and pivoting arrangement shall be designed so that a minimum of side thrust is transmitted to the valve stem seal when the lever is operated. Provision shall be made to ensure that the lever will operate properly after normal wear of the valve parts has occurred. The contour and position of the handle and lever shall be designed for easy operation with bare, gloved, or mittened hand and for holding in either throttled or full-open position without undue fatigue of the operator.

3.7.4 Spout. The nozzle shall be provided with a spout conforming to Figure 3. The spout shall be provided with a knurled surface for hand-tighting into the nozzle body. The spout shall be anodized and the threaded surfaces treated with solid-film lubricant in accordance with 3.3.2. Spout dimensions shall conform strictly to the dimensions and tolerances shown in Figure 3.

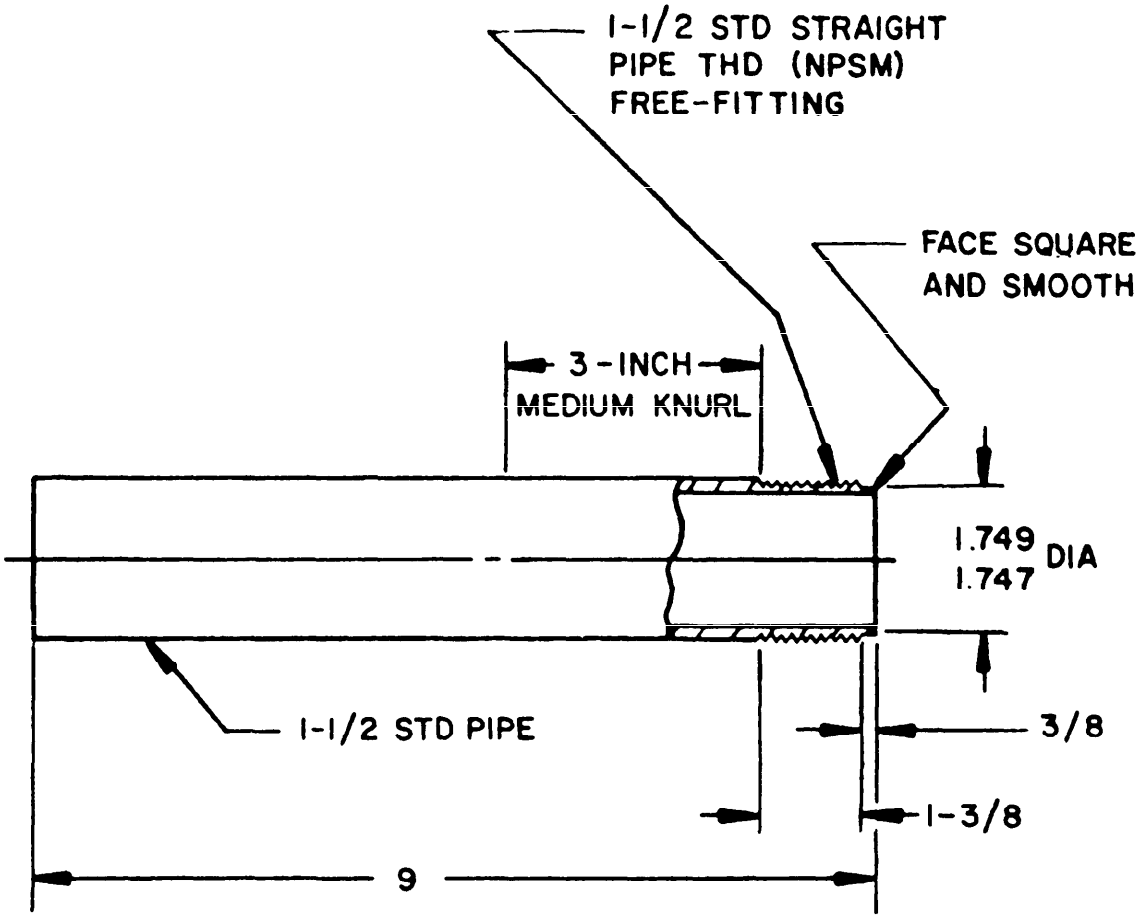
3.7.5 Strainers and strainer seals. Each nozzle shall be provided a strainer. Strainers shall conform to the dimensions shown in Figure 4 and to the material requirements of 3.3.1. Each strainer shall consist of an inner 100-mesh screen, an outer 6-mesh screen, and a ring fabricated and assembled in strict conformance to the dimensional tolerances shown in Figure 4. The screens shall be joined securely to the ring by soldering, brazing, spot welding, and/or heliarc welding in such a manner that all fluid flow is directed through the 100-mesh screen without any bypassing. Joining method and technique shall be selected to insure mechanical integrity and to minimize the amount of screen area blocked by joining material. Two standard size O-rings and one metal back-up ring shall be provided with each strainer and shall be installed as shown in Figure 2 to prevent fluid leakage between the spout and body and bypassing of the strainer.

3.7.6 Filler neck protector. The nozzle shall be provided with a removable, adjustable protector for aircraft filler necks, sized to fit the nozzle spout. The protector shall conform to Figure 5 and shall provide a rest for the nozzle during the refueling of thinwing aircraft in order to prevent wing-tank internal damage and wing skin damage. The filler neck protector shall not be displaced from a selected position on the nozzle spout when subjected to a sustained force of 25 pounds plus dead weight of the nozzle.

3.7.7 Dust cap. Each nozzle shall be provided with a dust cap corresponding in size to the spout. The dust cap shall consist of a close-fitting, removable, molded rubber cap at least 3 inches in length conforming to the general configuration shown in Figure 1. Dust cap shall be molded from rubber conforming to MIL-G-1086, Type II. The cap shall be provided with noncontinuous internal circular corrugations to provide the necessary retention on the end of the spout. The cap shall be fixed securely on the jumper (ground connections) cable at the location shown in Figure 1; the cap cable attachment and cap shall be of sufficient strength that the cap can be removed by pulling on the cable. The end of the cap shall be provided with a small orifice for breaking of vacuum within the cap caused by removal of cap from spout.

3.7.8 Electrostatic discharger. The nozzle shall be furnished with a electrostatic discharger jumper securely attached to the nozzle handle and dust cap in conformance with Figure 1. The electrostatic discharger assembly shall conform to MS25384-5 except that the cable length between the stud and plug shall be not less than 54 inches.

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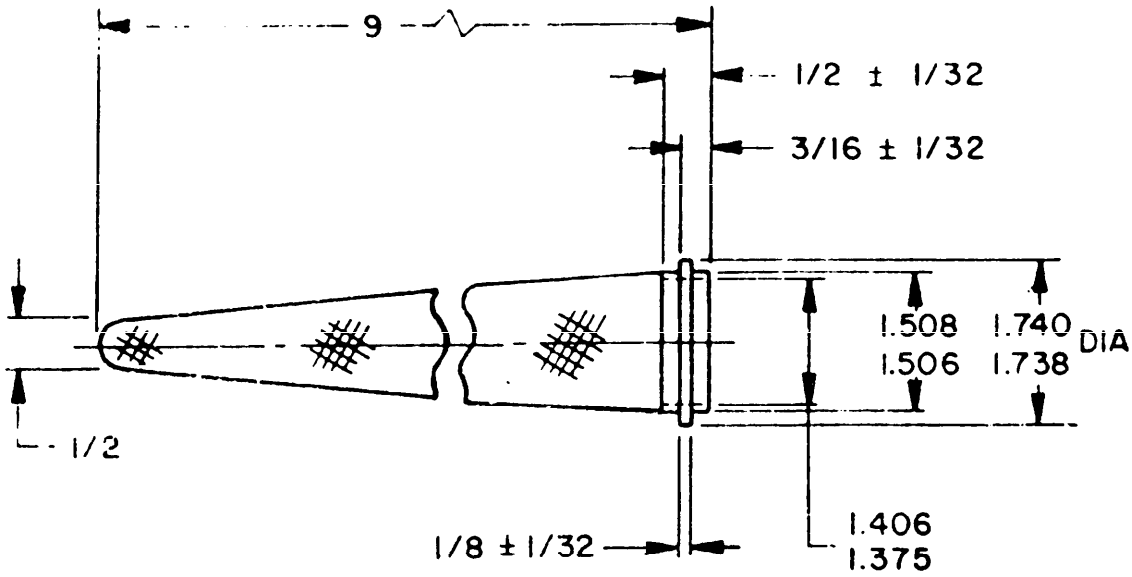


SPOUT MATERIAL:
6061-T6 ALUMINUM ROUND
PIPE, SCHEDULE 40
DIMENSIONS IN INCHES:
TOLERANCES: FRACTIONS ± 1/32
DEGREES ± 1°
BREAK-SHARP EDGES

NOZZLE SPOUT

FIGURE 3

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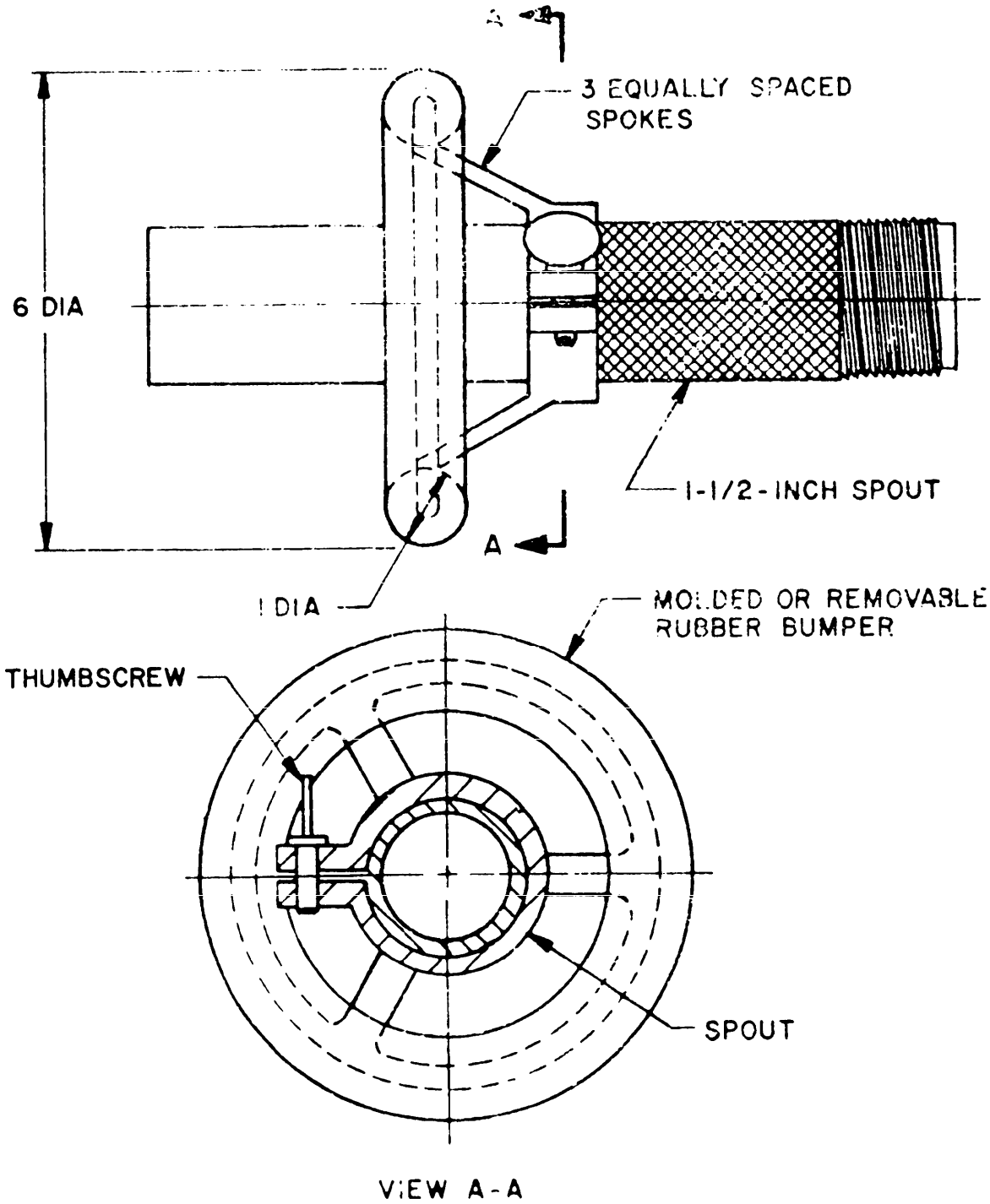
SQUARE-MESH WIRE CLOTH,
STANDARD MARKET GRADE:
OUTER: 6X6 WITH 0.035 DIA WIRE
INNER: 100X100 WITH 0.0045 DIA WIRE

DIMENSIONS IN INCHES TOLERANCE: FRACTIONS ± 1/16

NOZZLE STRAINER

FIGURE 4

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VIEW A-A

DIMENSIONS IN INCHES TOLERANCE FRACTIONS ± 1/8

AIRCRAFT FILLER NECK PROTECTOR
FIGURE 5

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3.8 Weight. The weight of each complete nozzle, including all components listed in 3.2 herein, shall not be greater than 9 pounds.

3.9 Threads. All pipe and machine screw threads shall be in the inch system and shall conform to Handbook H-28.

3.10 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The manufacturer's part number and drawing number shall be the same.

3.11 Identification of product. All major components (see 3.2) except electrostatic discharger shall be marked in accordance with MIL-STD-130, and all other components and parts shall be so marked whenever feasible.

3.12 Workmanship.

3.12.1 General. Nozzles shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

3.12.2 Dimensions. Size and dimensions shall be in accordance with the requirements specified herein and with Figure 1 through 5. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practice. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the nozzles, they shall be held or limited accordingly.

3.12.3 Finish. Machine surfaces shall be finished in accordance with best shop practice for the intended end-use. Cast surfaces shall be maintained as smooth as possible, consistent with the best foundry practice. All burrs and sharp edges shall be removed from all components and parts.

3.12.4 Cleaning. All castings shall be cleaned by appropriate methods to remove sand and other foreign material prior to subsequent fabrication and treatment. All components shall be cleaned prior to assembly to remove all oil, grease, metal chips, and other foreign material. Each nozzle shall be checked during and after assembly and shall be cleaned as necessary to ensure freedom from foreign material. The finished nozzles shall be supplied in clean and dry condition.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of testing. Quality assurance testing shall be classified as follows:

Preproduction testing	See 4.3
Acceptance testing	See 4.4

4.3 Preproduction testing.

4.3.1 General. Preproduction testing shall consist of performance of all of the tests in 4.5.

4.3.2 Test samples. Preproduction test samples shall consist of two nozzles fabricated in the same facilities and using the same materials, components, and production technologies that will be used for the production lot of nozzles. One of the preproduction samples shall be submitted to the procuring activity in unused condition, and the other shall be subjected to the tests described herein to determine conformance to the requirements of this specification and then submitted to the procuring activity. Each preproduction nozzle shall be fully assembled and shall include all components listed in 3.2. At his own option and expense, the contractor may test more than one nozzle so as to reduce the test time; all nozzles tested shall be submitted to the procuring activity.

4.3.3 Test report. After completion of preproduction testing, the contractor shall submit to the procuring activity a test report prepared in accordance with MIL-STD-831 and shall be authenticated as provided by paragraphs 4.1.4.1 and 4.1.4.2 of MIL-STD-831. This report shall include, but not be limited to: (a) identification or description, or both, of all test facilities, equipment, and procedures, (b) test results, (c) 8 X 10-inch photographs of the nozzles before and after testing, plus other photographs showing test equipment or test results as applicable, (d) test equipment calibration data, (e) either oscilloscope photographs or oscillograms representing the results of testing performed in compliance with 4.5.5 plus calibration data, (f) statement of compliance with all material and process requirements of this specification, including specific listing of commercial equivalents where applicable, (g) certification, when applicable, of elastomer conforming to MIL-R-25897, (h) manufacturer's top (assembly) drawing of nozzle, (i) Cutaway drawing of the nozzle together with an explanation of how it functions. (j) flow pressure curve required by 4.5.3, (k) test fluid properties as required by 4.3.9, and (l) comments of contractor and test activity on test procedures and results, nozzle features, and specification requirements.

4.3.4 Rejection and retest. Failure to meet any one requirement in the preproduction testing shall cause rejection. In such a case, the contractor shall be responsible for correction of defects, resubmission of preproduction samples, and repeating the preproduction tests. The procuring activity reserves the right to require complete preproduction testing on the resubmitted samples but may waive the requirement for some of the repeat tests, provided the contractor submits evidence that correction of defects has not changed the performance of the test samples with respect to these particular tests.

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4.3.5 Sequence of testing. Preproduction tests on each test sample shall be conducted in the following order:

Examination of product	4.5.1
Leakage tests	4.5.2
Flow test	4.5.3
Lever pull test	4.5.4
Surge test	4.5.5
Low-temperature tests	4.5.6
Functional tests after low-temperature exposure	4.5.7
Endurance test	4.5.8
Salt fog test	4.5.9
Filler neck protector test	4.5.10
Impact test	4.5.11

Both preproduction test samples shall be subjected to all of these tests. Test methods shall be identical for the two preproduction test samples except in the case of the leakage test (4.5.2), in which one sample shall be tested with water and the other with the test fluid specified in 4.5.

4.4 Acceptance testing.

4.4.1 Sampling plan A. A random selection of 10 percent of the production lot shall be taken and subjected to examination of product (4.5.1) and leakage tests conducted in accordance with 4.5.2, except that distilled or demineralized water shall be used as the test fluid, and the holding time at each pressure shall be 2 minutes. There shall be no leakage of test fluid through the valve in the test with valve closed, and no leakage of test fluid through any of the seals or other nozzle components during any of the leakage tests. After completing the leakage tests, the nozzle shall be dried by means of clean, dry air.

4.4.2 Sampling plan B. Two nozzles shall be selected at random from each 500 units of fraction chereof in the production lot and subjected to the impact test (4.5.11) followed by the endurance test (4.5.8). These nozzles shall not be returned to the production lot after test.

4.4.3 Rejection and retest. When one or more nozzles fail to meet any part of the acceptance test requirements, the acceptance of all nozzles shall be withheld. The cause of failure shall be determined and corrected. After corrections have been made on the entire production lot, it shall be resampled and retested in accordance with 4.4.1 and 4.4.2.

4.4.4 Defects in items already accepted. In the event of a test failure, the contractor shall advise the procuring activity in full detail on all defects that could exist in items already accepted and on methods of correcting such defects.

4.5 Test methods. Except as otherwise specified herein, test fluid shall consist of mineral spirits conforming to TT-T-291, grade 1 to which has been added 0.10 to 0.15 percent by volume of anti-icing additive conforming to MIL-I-27686. Except as otherwise specified herein, all tests shall be conducted at normal room temperature.

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4.5.1 Examination of product. Each test sample shall be inspected to determine compliance with the requirements specified in Section 3 with respect to workmanship, dimensions, weight, and material.

4.5.2 Leakage tests.

4.5.2.1 Leakage test with valve closed. The nozzle inlet shall be attached to any suitable system supplying test fluid under pressure. With a fluid supply pressure of not more than 2 psig, the operating lever shall be actuated to open and close the valve two times in such a manner that the nozzle is filled completely with test fluid, after which the static pressure shall be adjusted to 2 psig and maintained for 5 minutes with the valve closed. The static pressure shall then be increased and held successively at 50 and 100 psig for 5 minutes each. There shall be no leakage of test fluid through the valve, stem seal, or any component of the nozzle.

4.5.2.2 Leakage test with valve open. The spout shall be loosened and retightened by hand without any abnormal force, and shall not be further tightened during test. The spout shall then be closed or plugged by any suitable means. With the valve open, fluid pressure of 5 psig shall be applied to the nozzle inlet and maintained for 5 minutes. There shall be no leakage of test fluid through the stem seal, spout-strainer seals, or any component of the nozzle.

4.5.3 Flow test. The nozzle inlet shall be attached to a straight length of 2-inch pipe with a pressure gage or transducer located in the pipe at a distance of 21 inches from the nozzle inlet. Provision shall be made for measuring the temperature of the test fluid in this pipe. The pipe shall be attached to a pumping system of the required capacity, equipped with a suitable flow meter, and equipped for heating or cooling the fluid if necessary. With the nozzle valve in wide-open position, fluid shall be pumped through the nozzle at 135 ±5gpm for a minimum of 5 minutes while maintaining the fluid temperature between 70 and 90°F. Pumping shall be continued as long as necessary to establish steady-state conditions. At this time, the fluid flow rate, pressure, and temperature shall be recorded and reported. Additional tests shall be made in a similar manner at flow rates of approximately 40, 80, and 200 gpm, and the relationship shall be reported in the form of a flow-pressure curve showing the fluid temperature corresponding to each data point. No correction for fluid properties shall be applied to the flow-pressure data; however, the test fluid density and kinematic viscosity at the average test temperature shall be reported. The nozzle inlet pressure shall not be greater than 25 psig at a flow rate of 135 gpm. There shall be no leakage of the nozzle during this test.

4.5.4 Lever pull. With the nozzle mounted as specified in 4.5.3, the system shall be adjusted to furnish a static pressure of 50 psig at the nozzle inlet with the nozzle valve closed. The operating lever shall be actuated by applying a force at a point on the lever corresponding to the point farthest from the pivot that is covered by a normal hand in normal operating-grip position with the valve closed. (For the purposes of determining this point, a normal hand shall be defined as one measuring 3-1/4 inches across the second finger-joints.) This point shall be determined for the nozzle being tested and shall

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be further defined and reported for this particular nozzle in terms of distance from the nearest pivot point or fulcrum of the operating lever. The force shall be applied and measured in a direction perpendicular to the following: (a) maximum "breakaway" force required to open valve against 50-psig static pressure, (b) force required to hold valve in "cracked" position with trickle flow at 50-psig inlet pressure, (c) force required to hold valve in throttling position approximately one-quarter open at 50 psig inlet pressure, and (d) force required to hold valve full open at a flow rate of 135 gpm with inlet pressure as required to give flow rate. The pumping system shall be readjusted during the course of making these measurements as required to maintain the specified inlet pressures under flow conditions. The force required shall not exceed 15 pounds under any of these conditions.

4.5.5 Surge test. The nozzle shall be mounted as specified in 4.5.3 and attached to a suitable pumping system by means of a 50-foot length of 2-inch standard commercial collapsible fueling hose. With the nozzle valve wide open and the fluid flow rate set at 135 ±5gpm, the operating lever shall be released suddenly to permit the valve to close. Electronic recording and indicating equipment shall be used to determine and record the initial full-flow pressure, and the valve closing time (interval between handle release and return of pressure to the static shutoff valve). Oscilloscope photographs or oscillograms shall be included in the test report. The surge pressure (peak pressure minus statis shutoff pressure) as measured by the transducer 21 inches from the nozzle inlet shall not be greater than 50 psi, and the valve closing time shall not be greater than 1.5 second. The valve shall shut off the flow of test fluid smoothly and completely and there shall be no leakage after shutoff.

4.5.6 Low-temperature tests. The nozzle, together with the required test equipment and test fluid, shall be placed in a low-temperature chamber and maintained at an ambient temperature of -65 ±2°F for 48 hours. While still being held at this ambient temperature, the nozzle shall then be subjected to the leakage and lever pull tests specified in 4.5.6.1, 4.5.6.2. Following these tests, the nozzle, equipment, and test fluid shall be maintained at an ambient temperature of -65 ±2F for an additional 48 hours, and then the low-temperature flow test specified in 4.5.6.3 shall be performed.

4.5.6.1 Low-temperature leakage test. The leakage test specified in 4.5.2.1 shall be performed at -65°F. There shall be no leakage of test fluid through the valve, stem seal, or any component of the nozzle.

4.5.6.2 Low-temperature lever pull test. The lever pull test specified in 4.5.10 shall be performed at -65°F. This test shall be performed as rapidly as possible, and the nozzle fluid inlet temperature shall not be above -55°F at the conclusion of the test. The force required shall not exceed 15 pounds under any of the conditions specified in 4.5.4.

4.5.6.3 Low-temperature flow test. The nozzle test fluid and apparatus specified in 4.5.3 shall be placed in a low-temperature chamber and maintained at an ambient temperature of -65 ±2F for 48 hours. Immediately following this cold-soad period, the pump shall be turned on with the nozzle valve wide open, and the system shall be adjusted to maintain a nozzle inlet pressure of 25 ±1 psig. The nozzle inlet fluid pressure and temperature and the flow rate shall

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be recorded at one-minute intervals. The test shall be continued until the measured flow rate has reached at least 135 gpm. If the initial measured flow rate is less than 135 gpm, pumping shall be continued, allowing the fluid to warm up because of the pumping, and readings of pressure, fluid temperature, and flow-rate shall be continued at one-minute intervals until the flow rate is not less than 135 gpm. The nozzle inlet fluid temperature shall not be higher than -55°F at the time the flow rate has reached 135 gpm. There shall be no leakage of the nozzle during this test.

4.5.7 Functional tests after low-temperature exposure. Following the low-temperature tests specified in 4.5.6, the nozzle shall be subjected to leakage, flow, and lever pull tests at room temperature in accordance with 4.5.2, 4.5.3 and 4.5.4 and shall meet the requirements listed herein.

4.5.8 Endurance test. The nozzle shall be installed in a flow system, and test fluid shall be passed through the nozzle at a flow rate of at least 50 gpm, measured with the nozzle valve side open. This wide-open flow rate shall be maintained at 50 gpm or greater throughout the course of the test. The valve shall be fully opened and closed 5000 times. The nozzle inlet fluid temperature, nozzle inlet pressures with valve in closed and wide-open positions, and flow rate with valve in wide-open position shall be recorded and reported at the start and end of the test. The valve shall function normally throughout the test. At the conclusion of the 5000 operating cycles, the nozzle shall be subjected to the leakage tests as specified in 4.5.2. There shall be no leakage of test fluid through the valve, stem seal, or any component of the nozzle. The nozzles shall then be disassembled and examined for wear, misalignment, "set" of elastomers and seals, and general condition. Any evidence of abnormal wear or malfunction shall be cause for rejection.

4.5.9 Salt fog test. The nozzle, including all components, with the dust cap installed on the spout and with the inlet connection plugged, shall be exposed to salt fog for 48 hours in accordance with Method 509 of MIL-STD-810. After this exposure, the nozzle shall be examined visually. Any loose corrosion products shall be removed by wiping or brushing after recording their presence. There shall be no pitting, etching, or localized corrosion as determined by visual examination without magnification. There shall be no rusting of plated bolts, nuts, or miscellaneous small parts other than minor edge rusting that does not interfere with the proper functioning of such parts. The operating lever and valve stem shall move freely, without any binding or restriction of motion in any position. The spout shall be removable by hand without application of abnormal force. There shall be no corrosion that will interfere in any way with normal functioning of the nozzle. All corrosion shall be reported in terms of location, type, extent, and severity, with photographs as needed. Rusting of the steel cable or corrosion of the electrostatic discharger hardware shall not be cause for rejection.

4.5.10 Filler neck protector test. The filler neck protector shall be positioned and finger tightened on the spout in three different and random locations on the spout. While in each of the three positions, the filler neck protector shall be subjected to a sustained load of 25 pounds plus the dead weight of the nozzle uniformly distributed on the rim of the protector. The filler neck protector shall not be displaced from its selected location on the spout due to this loading.

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4.5.11 Impact test. The nozzle shall include all components except the filler neck protector and dust cap. The dust cap shall be removed from the grounding cable. The grounding cable shall be wound around the spout, and the grounding plug taped to the side of the spout. The spout shall be installed hand-tight in the nozzle body. The nozzle shall be attached to one end of a 20-foot (minimum) length of 2-inch standard commercial collapsible fueling hose. The other end of the hose shall be attached to a locking-type swivel connection. The hose and nozzle shall be filled with water and pressurized to 1 and 2 psig while held at floor level. The nozzle shall be raised to a height of 12 feet and dropped directly onto a concrete floor, without any catch pan or other device on the floor that might reduce or modify the nature of the impact. The drop shall be performed 6 times, rotating the fixed end of the hose between drops so as to cause the nozzle to strike in a different position each time. The hose and nozzle shall be repressurized each time if the repositioning or the previous impact have caused loss of pressure. The total amount of leakage from the nozzle during this test shall not exceed one quart, determined approximately by mop-up and weighing. At the conclusion of the test the nozzle shall be examined to determine the extent of deformation, breakage, or other damage, and all visible damage shall be recorded descriptively and in photographs. Damage to the spout shall not be cause for rejection. There shall be no breaks or excessive deformation of any other component or any damage that interfers with normal functioning of the nozzle. The operating lever and valve stem shall move freely through full travel without any binding or difficulty in operation. The leakage test specified in 4.5.2 shall be performed; there shall be no leakage through the valve, stemseal, or any component of the nozzle.

5. PACKAGING

5.1 Preservation, packaging and packing. Nozzles shall be prepared for shipment in accordance with levels A, B or C of MIL-H-775 as specified (see 6.2).

6. NOTES

6.1 Intended use. Type MD-3 nozzles are intended for use on ground equipment for servicing aviation gasoline, jet fuel and water-alcohol mixtures (ADI) to aircraft. Type MD-3 nozzles can also be used for servicing automotive gasoline, diesel fuel, and light fuel oils to ground vehicles and ground equipment. The type MD-3 nozzle replaces both the MD-1 and MD-2 nozzles covered by MIL-N-26978A.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging, and packing required (see 5.1).

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6.3 Reclaimed materiels. The use of reclaimed materials shall be encouraged to the maximum extent possible.

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